

Started on Wednesday, 13 November 2024, 2:26 PM

State Finished

Completed on Wednesday, 13 November 2024, 2:56 PM

Time taken 30 mins 1 sec

Marks 2.00/3.00

Grade 2.00 out of 3.00 (66.68%)

Information

Information

This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade**.

The following rules apply:

- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

Question 1

Incorrect

Mark 0.00 out of 1.00

Find the solution to the initial value problem

$$\begin{cases} a_n - 2 \cdot a_{n-1} - 8 \cdot a_{n-2} = 0 \\ a_0 = 1 \\ a_1 = -8 \end{cases}$$

$a_n =$

Input your answer as a function of n . Any other variable is treated as a constant. E.g. if you think the answer is $2^n + 3$, you should enter **2^n + 3**.

Your last answer was interpreted as follows:

$$2^n$$

The variables found in your answer were: $[n]$

Question 2

Correct

Mark 1.00 out of 1.00

In this problem you are tasked with computing with elements in the symmetric group of permutations, $(S_5, \cdot)$.

Syntax guide: The answer to each task is an element of the group S_5 and should be given as a 2×5 matrix. E.g. if you think the answer to a question is the permutation denoted by

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 5 & 4 \end{pmatrix}$$

then you should answer

1	2	3	4	5
2	3	1	5	4

a)

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 5 & 4 & 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 2 & 1 & 4 & 3 \end{bmatrix} = \left| \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ 1 & 3 & 4 & 5 & 2 \end{array} \right|$$

Your last answer was interpreted as follows:

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 3 & 4 & 5 & 2 \end{bmatrix}$$

b)

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 3 & 4 & 2 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 4 & 2 & 1 & 3 \end{bmatrix} = \left| \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 1 & 4 & 5 \end{array} \right|$$

Your last answer was interpreted as follows:

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 1 & 4 & 5 \end{bmatrix}$$

c)

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 5 & 4 & 1 & 2 \end{bmatrix}^{-1} = \left| \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ 4 & 5 & 1 & 3 & 2 \end{array} \right|$$

Your last answer was interpreted as follows:

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 5 & 1 & 3 & 2 \end{bmatrix}$$

d)

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 2 & 1 & 4 & 3 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 5 & 4 & 1 \end{bmatrix}$$

Your last answer was interpreted as follows:

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 5 & 4 & 1 \end{bmatrix}$$

Question **3**

Correct

Mark 1.00 out of 1.00

a) Let $G = (\mathbb{Z}/2)^{\times 4}$, and let $H \subset G$ be the subgroup

$$H = \{[0, 0, 0, 0], [0, 0, 1, 1], [0, 1, 0, 0], [0, 1, 1, 1], [1, 0, 0, 0], [1, 0, 1, 1], [1, 1, 0, 0], [1, 1, 1, 1]\}.$$

What is the index of $H \subset G$?

$$[G : H] = 2$$

Your last answer was interpreted as follows:

$$2$$

b) Let D_4 be the dihedral group of order 8 of symmetries of the unit square centered at the origin of $\mathbb{R}^2$. Let $\sigma \in D_4$ be the symmetry that reflects around the line $(y=-x)$.

Compute the index of the subgroup generate by σ :

$$[D_4 : \langle \sigma \rangle] = 4$$

Your last answer was interpreted as follows:

$$4$$

Question **4**

Correct

Mark 0.00 out of 0.00

Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.

Tests missing this signing code will be ignored and will not count towards the final score.

Key: 277

Signing code:

Your last answer was interpreted as follows:

31647

[◀ Test 2 \(topics 4-6: Languages, Machines, Regular expressions, Correct programs, Relations\)](#)

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